

Infinitesimal Calculus 2, Moed A

2022, Reichman university Solutions

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Instructions:

1. Duration: 3 hours.
2. Only a **basic** calculator and the attached formula sheets are permitted in the exam.

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The following exam has 5 questions. You must answer exactly 4 questions. Each question is worth 25 credits.

You must explain in full details all of your answers in all parts of the exam, and state all of the conditions for every theorem and/or claim. **Unexplained answers will not receive any credit. It is forbidden to rely on any material that was not learned in the course. You must write your solutions only in the designated areas. Do not write at the back of the papers!**

1. (25 points)

(a) (13 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Suppose that the series $\sum_{n=1}^{\infty} a_n$ is convergent.

i. (7 points) Prove that $\lim_{n \rightarrow \infty} a_n = 0$.

ii. (6 points) Prove that the series $\sum_{n=1}^{\infty} \frac{a_n}{n^2}$ converges absolutely.

Solution: We have $\lim_{n \rightarrow \infty} \left[\frac{|a_n|}{n^2} \right] = \lim_{n \rightarrow \infty} |a_n| = 0$, and as $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent then $\sum_{n=1}^{\infty} \frac{|a_n|}{n^2}$ is convergent, that is, the series $\sum_{n=1}^{\infty} \frac{a_n}{n^2}$ converges absolutely.

(b) (12 points) Determine whether the following improper integral is convergent or divergent: $\int_0^1 \frac{e^x + x}{\sqrt{e^x - 1}} dx$.

Solution:

Note that $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$, which implies that there exists a $0 < \delta_1 < 1$ for which for every $0 < x < \delta_1$ we have $\frac{e^x - 1}{x} > \frac{1}{2}$.

Thus,

$$\int_0^1 \frac{e^x + x}{\sqrt{e^x - 1}} dx = \int_0^{\delta_1} \frac{e^x + x}{\sqrt{e^x - 1}} dx + \int_{\delta_1}^1 \frac{e^x + x}{\sqrt{e^x - 1}} dx$$

Note that the second integral is finite, as the function $\frac{e^x + x}{\sqrt{e^x - 1}}$ is continuous on $[\delta_1, 1]$ and therefore integrable. As for the first integral,

$$\int_0^{\delta_1} \frac{e^x + x}{\sqrt{e^x - 1}} dx = \int_0^{\delta_1} \frac{e^x + x}{\sqrt{x} \sqrt{\frac{e^x - 1}{x}}} dx \leq \int_0^{\delta_1} \frac{2(e^x + x)}{\sqrt{x}} dx \leq \int_0^{\delta_1} \frac{2(e + 1)dx}{\sqrt{x}} = 2(e + 1) \int_0^{\delta_1} \frac{dx}{\sqrt{x}} \leq 2(e + 1) \int_0^1 \frac{dx}{\sqrt{x}} < \infty$$

which implies that $\int_0^1 \frac{e^x + x}{\sqrt{e^x - 1}} dx < \infty$.

We now give an alternative solution:

Note that the function $\frac{e^x + x}{\sqrt{e^x - 1}}$ is nonnegative for every $x \in (0, 1]$. As $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$, then $e^x \geq 1 + x$ for every $x \geq 0$ and thus,

$$\int_0^1 \frac{e^x + x}{\sqrt{e^x - 1}} dx \leq \int_0^1 \frac{e + 1}{\sqrt{e^x - 1}} dx \leq \int_0^1 \frac{4}{\sqrt{e^x - 1}} dx \leq \int_0^1 \frac{4}{\sqrt{x}} dx < \infty$$

implying that $\int_0^1 \frac{e^x + x}{\sqrt{e^x - 1}} dx < \infty$.

2. (25 points) Let $x_0 \in \mathbb{R}$ and let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers.

(a) (15 points) Let $x_1, x_2 \in \mathbb{R}$ such that $|x_2 - x_0| < |x_1 - x_0|$. Suppose that the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges at $x = x_1$.

Prove that the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ converges absolutely at $x = x_2$.

(b) (3 points) Define the term: The convergence radius of the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$. Note: you do not have to define the term: a convergent series.

Solution: The convergence radius of the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$ is defined as

$$R = \sup \left\{ r \geq 0 : \sum_{n=0}^{\infty} a_n(x - x_0)^n \text{ is convergent for every } x \in [x_0 - r, x_0 + r] \right\}$$

(c) (7 points) Determine whether the following series is convergent or divergent: $\frac{1}{\sqrt{2}-1} - \frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}-1} - \frac{1}{\sqrt{3}+1} + \dots$

Solution: ATC that there exists a real number $s \in \mathbb{R}$ such that $s = \frac{1}{\sqrt{2}-1} - \frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}-1} - \frac{1}{\sqrt{3}+1} + \dots$. By the Parentheses test, we have

$$s = \left(\frac{1}{\sqrt{2}-1} - \frac{1}{\sqrt{2}+1} \right) + \left(\frac{1}{\sqrt{3}-1} - \frac{1}{\sqrt{3}+1} \right) + \dots = \frac{2}{1} + \frac{2}{2} + \frac{2}{3} + \dots = \sum_{n=1}^{\infty} \frac{2}{n} = 2 \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

Contradiction.

3. (25 points)

(a) (19 points)

- i. (3 points) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function that is twice differentiable at $x_0 = 0$. Define the terms: the Taylor polynomial and the Taylor remainder of order 2 of f about 0.

Solution: The Taylor polynomial of order 2 of f about 0 is defined as

$$T_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2.$$

The Taylor remainder of order 2 of f about 0 is defined as $R_2(x) = f(x) - T_2(x)$ for every x in a neighbourhood of 0.

- ii. (4 points) Consider the function defined by $f(x) = \sqrt{1+x}$ for every $x > -1$. Compute the Taylor polynomial of f of order 2 about $x_0 = 0$.

Solution: We have $f(x) = \sqrt{1+x} \Rightarrow f(0) = 1; f'(x) = \frac{1}{2\sqrt{1+x}} = \frac{1}{2}(1+x)^{-\frac{1}{2}} \Rightarrow f'(0) = \frac{1}{2}; f''(x) = -\frac{1}{4}(1+x)^{-\frac{3}{2}} \Rightarrow f''(0) = -\frac{1}{4}$.

It follows that

$$T_2(x) = f(0) + f'(0)x + \frac{f''(0)}{2}x^2 = 1 + \frac{x}{2} - \frac{x^2}{8}$$

- iii. (12 points) Prove that for every $x \in \mathbb{R}$ we have

$$1 + \frac{1}{2}x^4 - \frac{1}{8}x^8 \leq \sqrt{1+x^4} \leq 1 + \frac{1}{2}x^4 - \frac{1}{8}x^8 + \frac{1}{16}x^{12}$$

Solution: We show that for every $0 \leq x$ we have

$$1 + \frac{1}{2}x - \frac{1}{8}x^2 \leq \sqrt{1+x} \leq 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3$$

and then, substituting x^4 into the above inequality yields the required result.

First, if $x = 0$ then we clearly have an equality. Else, $0 < x$ and by Taylor's remainder theorem, there exists a point $0 < c < x$ for which

$$R_2(x) = \frac{f'''(c)}{3!}x^3$$

Note that $f'''(x) = \frac{3}{8}(1+x)^{-\frac{5}{2}}$, and as $0 < c < x$ we have $R_2(x) \geq 0$ and

$$0 \leq \sqrt{1+x} - T_2(x) = R_2(x) = \frac{3}{8 \cdot 6(1+c)^{\frac{5}{2}}}x^3 \leq \frac{3}{48}x^3 = \frac{x^3}{16}$$

Adding $T_2(x)$ to both sides yields the result.

- (b) (6 points) Prove or disprove the following statement: we have

$$1 - \frac{\pi^2}{3!} + \frac{\pi^4}{5!} - \frac{\pi^6}{7!} + \dots = 0.$$

Solution: We have for every $x \in \mathbb{R}$, $\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$. Thus

$$0 = \sin(\pi) = \pi - \frac{\pi^3}{3!} + \frac{\pi^5}{5!} - \dots = \pi \left[1 - \frac{\pi^2}{3!} + \frac{\pi^4}{5!} - \frac{\pi^6}{7!} + \dots \right]$$

Dividing both sides by π gives the result.

4. (25 points)

(a) (19 points) Let $(a_n)_{n=1}^{\infty}, (b_n)_{n=1}^{\infty}$ be two sequences of real numbers. Suppose that $a_n, b_n \in \mathbb{N}$ for every $n \in \mathbb{N}$. Suppose also that the sequence $(b_n)_{n=1}^{\infty}$ is strictly increasing.

i. (4 points) Prove that $\lim_{n \rightarrow \infty} b_n = \infty$.

Solution: We prove by induction that $b_n \geq n$ for every $n \in \mathbb{N}$, implying that $\lim_{n \rightarrow \infty} b_n = \infty$ by Toast theorem. Clearly, as $b_1 \in \mathbb{N}$ then $b_1 \geq 1$. Now, suppose that $b_n \geq n$. As $b_{n+1} > b_n$, and $b_n, b_{n+1} \in \mathbb{N}$, then $b_{n+1} \geq b_n + 1 \geq n + 1$.

ii. (15 points) Suppose, in addition, that for every $2 \leq n \in \mathbb{N}$ we have $a_n \cdot b_{n-1} - a_{n-1} \cdot b_n = (-1)^n$. Prove that $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ exists.

Solution: For every $2 \leq n \in \mathbb{N}$ we have

$$\frac{a_n}{b_n} - \frac{a_{n-1}}{b_{n-1}} = \frac{a_n \cdot b_{n-1} - a_{n-1} \cdot b_n}{b_n b_{n-1}} = \frac{(-1)^n}{b_n b_{n-1}}$$

Thus, being a Telescopic sum, we have

$$\frac{a_N}{b_N} - \frac{a_1}{b_1} = \sum_{n=2}^N \left(\frac{a_n}{b_n} - \frac{a_{n-1}}{b_{n-1}} \right) = \sum_{n=2}^N \frac{(-1)^n}{b_n b_{n-1}}$$

Note that as $b_n \cdot b_{n-1}$ is strictly increasing and $\lim_{n \rightarrow \infty} (b_n \cdot b_{n-1}) = \infty$, then $\sum_{n=2}^{\infty} \frac{(-1)^n}{b_n b_{n-1}}$ is a Leibniz series, which implies that $\exists s \in \mathbb{R}$ such that $s = \sum_{n=2}^{\infty} \frac{(-1)^n}{b_n b_{n-1}}$. Thus, $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ exists and

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{a_1}{b_1} + s.$$

(b) (6 points) Compute the following limit, or prove that it does not exist:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^3}{x^4 + 2y^8}$$

Solution: We have

$$\lim_{(x,y) \rightarrow (0,0), x=y^2} \frac{x^2 y^3}{x^4 + 2y^8} = \lim_{y \rightarrow 0} \frac{y^4 y^3}{y^8 + 2y^8} = \lim_{y \rightarrow 0} \frac{y^7}{3y^8} = \lim_{y \rightarrow 0} \frac{1}{3y} = \text{DNE}$$

thus, the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^3}{x^4 + 2y^8}$ does not exist.

5. (25 points)

(a) (16 points) Consider the power series $\sum_{n=1}^{\infty} \frac{nx^n}{(n-1)!}$.

i. (4 points) Find the range of convergence of the above power series.

Solution: Let $0 \leq R \leq \infty$ be the convergence radius of the power series. Then

$$L = \lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} = \lim_{n \rightarrow \infty} \frac{(n+1)(n-1)!}{n \cdot n!} = \lim_{n \rightarrow \infty} \frac{n+1}{n^2} = \lim_{n \rightarrow \infty} \frac{1 + \frac{1}{n}}{n} = 0$$

Thus, $R = \infty$, and the range of convergence of the power series is $\mathbb{R}$.

ii. (12 points) Consider the function defined by $f(x) = \sum_{n=1}^{\infty} \frac{nx^n}{(n-1)!}$ for every x in the range of convergence that you found in item (i). Find a simple expression (without a sigma) for f on its domain, and conclude the value of the series $\sum_{n=0}^{\infty} \frac{n^2}{n!}$.

Solution: Using term-by-term differentiation, then for every $x \in \mathbb{R}$ we have

$$f(x) = \sum_{n=0}^{\infty} \frac{n^2 x^n}{n!} = \sum_{n=1}^{\infty} \frac{n^2 x^n}{n!} = \sum_{n=1}^{\infty} \frac{nx^n}{(n-1)!} = x \sum_{n=1}^{\infty} \frac{nx^{n-1}}{(n-1)!} = x \sum_{n=1}^{\infty} \frac{d}{dx} \frac{x^n}{(n-1)!} = x \frac{d}{dx} \sum_{n=1}^{\infty} \frac{x^n}{(n-1)!}$$

Note that

$$\sum_{n=1}^{\infty} \frac{x^n}{(n-1)!} = x \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!} = x \sum_{n=0}^{\infty} \frac{x^n}{n!} = xe^x$$

It follows that $f(x) = x \cdot (xe^x)' = x(e^x + xe^x) = x^2e^x + xe^x$ for every $x \in \mathbb{R}$. Finally, $f(1) = \sum_{n=0}^{\infty} \frac{n^2}{n!} = 2e$.

- (b) (9 points) Let $a, b \in \mathbb{R}$ such that $a < b$ and let f be a function that is defined on $[a, b]$. Suppose that f^2 is integrable on $[a, b]$ and that $\int_a^b f^2(x) dx = 0$. Let $x_0 \in (a, b)$. Prove that if f is continuous at x_0 , then $f(x_0) = 0$.

Solution: Consider the function defined by $F(x) = \int_a^x f^2(t) dt$, for every $a \leq x \leq b$. Note that $0 \leq F(x) \leq \int_a^b f^2(x) dx$ for every $a \leq x \leq b$, implying that $F(x) = 0$ for every $x \in [a, b]$. As f is continuous at x_0 , then f^2 is continuous at x_0 . Thus, by the Fundamental theorem of calculus, $f^2(x_0) = F'(x_0) = 0$, which implies that $f(x_0) = 0$.

BEHATZLAHA!!!!